Chapter 6 Breakout (Pt. I)

STAT 4714

Questions

1. The daily amount of coffee (in liters) dispensed by a machine is a random variable X having a continuous uniform distribution with $A = 7$ and $B = 10$. Find the probability that on a given day, the amount of dispensed coffee will be

   (a) at most 8.8 liters.

   (b) between 7.4 and 9.5 liters.

$X \sim \text{Uniform}(A=7, B=10) \Leftrightarrow u(x; 7, 10)$

(a) $P(X < 8.8) = \int_7^{8.8} \frac{1}{10-7}\,dx = \frac{1}{3}(8.8-7)$

$= \underline{3/5}$

(b) $P(7.4 < X < 9.5)$

$= \int_{7.4}^{9.5} \frac{1}{10-7}\,dx = \frac{1}{3}(9.5-7.4)$

$= \underline{7/10}$

2. Given a standard normal distribution, find the value of k such that

(a) $P(Z > k) = 0.2946$

(b) $P(Z < k) = 0.0427$

(c) $P(-0.93 < Z < k) = 0.7235$

$n(x;0,1)$

(a) $P(Z > k) = 1 - P(Z \le k) = 0.2946$

$\Rightarrow P(Z \le k) = 0.7054$

$\Rightarrow \int_{-\infty}^{k} \frac{1}{\sqrt{2\pi}} \exp\{-\frac{1}{2}z^2\} dz = 0.7054$

$\Rightarrow k = \text{qnorm}(0.7054, 0, 1) = \underline{0.5400}$

(b) $P(Z < k) = 0.0427$

$\Rightarrow k = \text{qnorm}(0.0427, 0, 1) = \underline{-1.7202}$

(c) $P(-0.93 < Z < k) = 0.7235$

$\Rightarrow P(Z < k) - \underbrace{P(Z < -0.93)}_{\text{pnorm}(-0.93,0,1) = 0.1762} = 0.7235$

$\Rightarrow P(Z < k) = 0.8997 \Rightarrow \underline{k = 1.2798}$

3. Given the normally distributed variable X with mean 18 and standard deviation 2.5, find

   (a) $P(X < 15)$

   (b) the value of k such that $P(X > k) = 0.1814$

   (c) $P(17 < X < 21)$

(a) $P(X<15) = \int_{-\infty}^{15} \frac{1}{\sqrt{2\pi}(2.5)} \exp\left\{-\frac{1}{2(2.5)^2}(x-18)^2\right\} dx$

$= \text{pnorm}(15, 18, 2.5) = \underline{0.1151}$

(b) $P(X>k) = 1 - P(X \le k) = 0.1814$

$\Rightarrow P(X \le k) = 0.8186$

$\Rightarrow k = \text{qnorm}(0.8186, 18, 2.5) = \underline{20.2751}$

(c) $P(17<X<21) = P(X \le 21) - P(X \le 17)$

$= \text{pnorm}(21, 18, 2.5) - \text{pnorm}(17, 18, 2.5)$

$= \underline{0.5404}$

4. It's reported that mice will live an average of 40 months when their diets are sharply restricted and enriched with vitamins and proteins. Assuming that the lifetimes are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live

   (a) more than 32 months

   (b) between 37 and 49 months.

$\mu = 40, \sigma = 6.3 \qquad n(x; 40, 6.3)$

(a) $P(X > 32) = 1 - P(X \leq 32)$

$= 1 - \text{pnorm}(32, 40, 6.3) = \underline{0.8979}$

(b) $P(37 < X < 49) = P(X \leq 49) - P(X \leq 37)$

$= \text{pnorm}(49, 40, 6.3) - \text{pnorm}(37, 40, 6.3)$

$= \underline{0.6065}$

5. The average life of a certain type of small motor is 10 years with a standard deviation of 2 years. The manufacturer replaces all motors that fail while under guarantee. If the manufacturer is willing to replace only 3% of motors that fail, how long a guarantee should be offered? Assume the lifetime of the motors follow a normal distribution.

$\mu = 10 \quad \sigma = 2 \quad \Rightarrow n(x; 10, 2)$

goal: find value k such that 3% is less than k

$P(X < k) = 0.03$

$\Rightarrow k = \text{qnorm}(0.03, 10, 2) = \underline{6.2384 \text{ years}}$